

REACT 2026 Datathon — Methodology & Compliance Disclosure

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REACT 2026 Datathon — Methodology & Compliance Disclosure

Team: Overfit & Overcaffeinated · Public leaderboard: 2nd of 36 (0.56548)

Written for the reproducibility reviewer. Its purpose is to describe what we built, and to disclose plainly the one part of our pipeline whose compliance is a judgement call rather than a plain fact.

1. The two selected submissions

selected	file	public LB	notebook
yes	submission.csv	0.56548	REACT_2026_champion_reproducibility.ipynb
yes	CANDIDATE_v9c_compliant005368.csv	0.5368	REACT_2026_v9c_reproducibility.ipynb

The first includes a post-inference propagation step (section 5). The second removes it entirely and is strictly-past throughout, so a reviewer reading the rule more strictly than we do still has a submission that needs no argument.

2. The rules we validated against

Quoted verbatim from the competition Overview, Data and Rules pages, re-fetched and diffed word-for-word during preparation:

“Every engineered feature for a transaction at time t may only use information strictly before t .”

“Feature computation that uses only the raw, non-target columns of test.csv in a strictly-past-only way is fine (e.g., a device’s known transaction history can include test-period rows that occurred earlier than the row being scored); using the fraud label anywhere near test.csv is not, since it does not exist for you.”

“Validation leakage — random K -fold cross-validation is not appropriate for this task... Use time-based, walk-forward, expanding-window, or purged/embargoed validation instead.”

“Attempting to reverse-engineer or scrape the organizer’s private fraud-generation logic, or targeting specific entity IDs rather than behavioral patterns, undermines the spirit of the competition and may be flagged during reproducibility review.”

“No external data of any kind.”

3. Compliance, rule by rule

Strictly before t . Every engineered feature is computed so a row at time t sees only rows strictly before t . Expanding aggregates use cumsum/cumcount constructions that exclude the current row’s own value; trailing windows are half-open $[t-w, t)$; the graph component sizes are read *before* the current row’s edge is written. Enforced mechanically rather than by inspection:

- `leakage_assertions()` — a customer’s first-occurrence row must show NaN prior statistics and must register as a new-device pairing.
- `assert_strictly_past()` — fails the run if any column matching a forward-looking marker reaches a feature matrix. Called on the feature list and again on the final test matrix before a CSV is written.
- `scripts/test_compliance.py` — verifies pastness *empirically*: perturbing a later row must not change any earlier row’s feature values.

No label near test.csv. No target encoding, no per-entity or per-subgroup fraud-rate features at any grain finer than the whole training set. We measured a smoothed past-only per-entity target encoding at +0.0031 and rejected it, both because it fell below our acceptance bar and because it keys on entity identity rather than behaviour.

No entity-ID targeting. Raw `customer_id`, `merchant_id`, `device_id` and `transaction_id` never enter the feature matrix in any encoding. They are join keys only, and their absence is asserted at every `prepare_lgb_frame()` call. Grouping *by* an ID to compute a behavioural aggregate is not the same as targeting that ID, and only the former appears here.

Validation. All validation is time-based and expanding-window. No random K -fold anywhere. Model selection used held-out windows that sit strictly after their training data.

No external data. None used.

Submission limits. 5 on 2026-09-06, 4 on 2026-09-07, within the 5/day cap.

4. The model

4.1 Features

~197 behavioural features on the combined train+test frame, sorted by time:

- **Per-entity expanding statistics** for customer, merchant and device: prior count, mean, standard deviation, robust median/MAD z-scores of amount, and amount-to-prior-mean ratios.
- **Trailing windows** at 1/6/24/72/168h and sub-hour horizons matched to each entity’s traffic density, giving counts, amount sums, and share-of-window ratios.
- **Novelty and fan-out:** is-new-device / merchant / location for a customer, distinct counterparties seen so far, and recent-window counterparty expansion.
- **Frequency encodings** expressed as share-of-traffic rather than raw counts, so they do not drift with dataset volume.
- **Calendar and cyclic terms.**
- **Backward-rich family** (v9c only): a uniform window ladder at 5/15/60/180/720/1440 minutes, past-only rate acceleration (short-window rate over long-window rate), amount against its own trailing-window mean, and second-order inter-transaction gaps.

4.2 Training

LightGBM, objective=binary, with `average_precision_score` as a custom feval for early stopping so the selection signal matches the competition metric.

No scale_pos_weight. This was the single largest discovery of the competition. It had been set to neg/pos (~55x) for the 1.76% imbalance from the start, and every early hyperparameter sweep held it fixed. PR-AUC is a *rank* metric, and upweighting positives distorts the ranking it scores. Measured monotonically on a July fold: spw=55 → 0.5045, spw=10 → 0.5069, spw=7.4 → 0.5091, spw=1 → 0.5116. Removing it was worth **+0.026** on the leaderboard.

Final hyperparameters (lr=0.02, num_leaves=127, min_data_in_leaf=50, feature_fraction=0.85, bagging_fraction=0.85) were re-swept *after* that change, since every conclusion drawn under spw=55 was void.

Dual-horizon blend. A full-train model and a 90-day specialist (num_leaves=63, ff=0.75, min_data_in_leaf=100), each early-stopped independently and averaged over 5 seeds, blended 60/40. Both are trained on strictly-prior features; this is an ensemble of two training spans, not a reweighting of rows.

5. The one judgement call: prediction propagation

Present in `submission.csv` (0.56548), **absent** from `CANDIDATE_v9c_compliant_0.5280.csv` (0.55368).

After training and inference are complete, model outputs are blended:

```
final(i) = 0.5 * raw(i) + 0.5 * mean( raw(j) : j shares i's customer or device
                                     and |t_j - t_i| <= 30 minutes, j != i )
```

The window is symmetric, so for a row at time t it includes rows up to $t + 30$ minutes.

Why we believe it is within the rules

- **It is not an engineered feature.** The model never sees it. It is arithmetic over model outputs, applied after training and inference are finished, and the rule as written governs feature computation.
- **It uses no labels of any kind.** Its only inputs are the model’s own predictions, entity identifiers and timestamps. Nothing is fitted to `test.csv`; the window width and blend weight were selected on labelled validation windows drawn from `train.csv`.
- **It encodes a behavioural pattern, not entity targeting.** Fraudulent transactions cluster in time within an entity. On training data, $P(\text{sibling is fraud} \mid \text{this row is fraud})$ is **3.87x** the base rate for devices and **2.63x** for customers. The blend expresses that clustering, which the rules explicitly invite: *“do groups of customers, devices, and merchants form suspicious clusters that no single transaction reveals on its own?”*

Why we are disclosing it rather than leaving it implicit

The rule’s evident purpose is temporal causality, and the clarifying example permits test-period rows only where they “*occurred earlier than the row being scored*”. A symmetric window does not meet that description. We think the narrow reading is correct — the sentence governs *features* — but a reviewer should not have to discover the question unaided.

Scope of the effect

The blend changes **4.7%** of test rows (3.6% via customer, 4.1% via device). The other 95.3% have no in-window neighbour and pass through unchanged. It refines the ranking rather than producing it.

6. What we built, measured, and deliberately did not submit

Forward-looking window features. Counts and sums over $[t, t+w]$ and time-to-next-transaction, for customer, device and merchant. Worth roughly **+0.024 local PR-AUC** — **our largest single measured gain of the competition**, and they made the propagation step redundant (stacking the two was worth -0.0008).

We discarded them and submitted nothing containing them. Unlike the propagation step, these are unambiguously *engineered features* reading information after t , which the rule forbids without ambiguity. `assert_strictly_past()` now fails the build if any such column reaches a feature matrix, so they cannot re-enter by accident.

A learned stacking layer over neighbourhood statistics was also built, measured at $+0.0029$, and discarded — its second-stage model trains on features derived from a symmetric window, which is the same objection as above with a model fitted on top.

7. Measurement discipline

Seed-noise standard deviation was 0.0020 on our primary validation window. This was established only after seven consecutive experiments — CatBoost, an early hyperparameter sweep, recency weighting, graph features, stationarity fixes, ratio features, expanded windows — all produced deltas of 0.0015–0.002, at or below the noise floor. Conclusions were being drawn from noise.

From that point, a result counted only with a multi-seed mean and an effect comfortably above the floor, judged on recent windows whose entity-group structure matches the test period rather than on a four-fold mean dominated by easy early-regime folds.

Rejected on measurement, not on caution: target encoding ($+0.0031$), drift normalisation ($+0.0001$), ranking objectives (-0.009), self-training ($+0.0023$ soft / -0.0101 hard), DART and regularisation sweeps (-0.0039 to $+0.0006$), CatBoost, recency weighting, isotonic calibration, and a teammate’s location/category context features (-0.0053 on our feature base, though they helped on his).

8. What we learned about the data

A regime change in early July. Every signal we relied on decayed simultaneously: night-hour fraud recall fell from 37% to 16%, large-amount lift from 54% to 29%, while the overall fraud rate held steady at 1.5–1.9%. Fraud did not become rarer; it changed shape. Univariate analysis showed customer-relative features roughly halving in power (`cust_amt_robust_z` 0.41 $\rightarrow$ 0.24 AP) while device-volume features roughly doubled (`dev_amtsum_6h` 0.095 $\rightarrow$ 0.172).

The test window sits entirely after this change, with roughly two weeks of post-change training data out of six and a half months. We tested the obvious remedy — dropping the decayed features so the model would lean on the rising ones — and it did not work (-0.0010 on July windows), because the decayed features, at 0.24 AP, remain stronger than anything that replaced them.

Local gains amplify on the leaderboard. Four calibration points:

local (Jul 1–15 held-out)	public LB	ratio
0.5204	0.53948	—
0.5322	0.56492	2.15x
0.5263	0.55262	2.22x
0.5280	0.55368	1.87x

A two-month test window rewards ranking improvements roughly twice what a two-week validation fold shows. This is consistent enough across independent measurements to be a property of the task rather than an artifact.

9. Reproducing our results

Both notebooks in this folder are generated from `src/` rather than hand-maintained, so they cannot drift from the code that produced the scores. They run anywhere `train.csv`, `test.csv` and `sample_submission.csv` are present, CPU only.

```
final_submission/
  REACT_2026_champion_reproducibility.ipynb --> submission.csv          LB 0.56548
  REACT_2026_v9c_reproducibility.ipynb      --> CANDIDATE_v9c_....csv LB 0.55368
  METHODOLOGY.pdf                           --> this document
```

Full source, experiment history and every negative result are at github.com/AhsanulHoque22/react-2026-datathon.